MIL-B-20292A(MU)
3 April 1967
SUPERSEDING
MIL-B-20292
21 November 1951

MILITARY SPECIFICATION

BLANKS, ROTATING BAND, FOR PROJECTILES

1. SCOPE

1.1 Scope. This specification covers rotating band blanks for projectiles.

1.2 Classification.

1.2.1 Compositions and tempers. The material covered by this specification shall be furnished in the compositions shown in table I, and in the following tempers, as specified (see 6.1):

Tempers

Annealed
Hard (as fabricated)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 1395

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SPECIFICATION

MILITARY

MIL-C-3993 - Copper and Copper Base Alloy Mill Products, Preparation for Shipment and Storage of

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Manufacture. Material furnished under this specification shall be in the form of a solid ring, rolled, drawn, forged, machined, or extruded. Annealed blanks, unless bright annealed, shall be pickled, washed and dried.

3.2 Chemical composition. The material shall conform to the chemical requirements of table I.

Table I - Chemical composition, percent

Copper and Copper Alloy Number	Copper[1]	Iron, max.	Lead, max.	Zinc, max.	Nickel	Manganese, max.	Total impurities, max.
130	99.88 min.	-	0.004	-	0.050 max.	-	0.12
220	89-91	0.05	.05	Rem.	-	-	0.10
702	Rem.	0.10	.05	-	2.0-3.0	0.40	0.30[2]

[1]Silver counting as copper
[2]Elements not otherwise specified

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3.2.1 Analysis shall be made regularly only for the elements specified in table I. If, however, the presence of other elements is suspected, or indicated in the course of routine analysis, further analysis shall be made to determine that the total of other elements is not in excess of the limits specified.

3.3 Mechanical properties. The material shall conform to the mechanical requirements of table II.

Table II. Mechanical properties of annealed rotating band blanks

Copper and Copper Alloy Number	Inside diameter of band blank, inches	Tensile strength, psi, max.	Elongation, percent, min. in 1 in.	Elongation, percent, min. in 2 in.	Bending test degrees
130	Up to 1.875 incl.	35,000	40	-	180
	Over 1.875	35,000	-	40	180
220	Up to 1.875 incl.	40,000	40	-	180
	Over 1.875	40,000	-	35	180
702	Over 7.50	40,000	-	35	180

3.3.1 Grain size. The average grain size of annealed blanks, except those for bands which are to be assembled to shell by hot forging (800°F min.) shall be as indicated in table III.

Table III. Average grain size

Copper and Copper Alloy Number	Caliber	Average grain size (mm) Min.	Average grain size (mm) Max.
130	All	0.035	-
220	20 mm to 105 mm incl.	0.045	0.090
	Over 105 mm	0.045	-

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3.3.2 Bend properties. Annealed blanks in the condition as furnished and hard blanks after being fully annealed shall withstand the bend test prescribed 9n 4.5.2.3 without cracking on the outside of the bent portion.

3.4 Dimensions and permissible variations. The dimensions and permissible variations shall be as specified in the contract, purchase order or as on the applicable drawing.

3.5 Workmanship. All blanks shall be free from slivers, pipes, laps, cracks, laminations, grooves, seams, excessive oxidation, flaws, deep scratches and other defects which may render them unsuitable for the purpose intended. All blanks shall be of uniform diameters, thicknesses, and widths, and free from scale, chips, oil, grease and dirt.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of not more than 20,000 blanks of the same composition, temper and size manufactured by the same process, except that where such a lot would have a nominal calculated weight of less than 1,000 pounds, the size of the lot may be greater and shall consist of a sufficient number of blanks to have a maximum weight of 1,000 pounds.

4.3 Sampling. Unless otherwise specified, and when applicable, the sampling plan and procedures used by the inspector in the determination of the acceptability of the product submitted by a supplier for inspection will be in accordance with provisions of MIL-STD-105.

4.3.1 Sampling for chemical analysis. A sample for chemical analysis shall be taken from each lot. It shall consist of not less than 2 ounces of clean chips, drillings, or millings, obtained from two or more blanks taken from the lot.

4.3.2 Sampling for mechanical properties.

4.3.2.1 Tension test. Two tension test specimens shall be taken from each lot. Each specimen shall be taken from a different blank.

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4.3.2.2 Bend test. When specified in the contract or order, two bend test specimens shall be taken from each lot. Each specimen shall be taken from a different blank.

4.3.2.3 Hard blanks shall be fully annealed as follows before specimens for tension and bending tests are taken therefrom.

Copper and Copper Alloy No.	Annealing temperature °F
130	1150-1350
220	1125-1400

4.3.3 Sampling for grain size. At least two different blank specimens shall be taken from each lot for grain size determination.

4.3.4 Sampling for visual and dimensional examination. Unless otherwise specified in the contract or order, samples for visual and dimensional examination shall be obtained in accordance with MIL-STD-105, Inspection level II, AQL (acceptable quality level) = 4.0 percent defective.

4.4 Examination.

4.4.1 Visual and dimensional. Blanks selected in accordance with 4.3.4 shall be visually examined to determine compliance with the requirements for workmanship (see 3.1 and 3.5). Standard measuring equipment shall be used to determine compliance with the requirements of 3.4.

4.4.2 Preparation for shipment. Examination of the packing and marking for shipment shall be made for conformance to the requirements of section 5.

4.5 Test methods and procedures.

4.5.1 Tension test specimens. The tension test specimens shall be cut circumferentially from the blanks. The blank, or section thereof, shall be straightened cold by pressure without hammering. Test specimens of a blank of various sizes shall be as follows: For blanks of inside diameter less than 7.5 inches but not less than 1 inch, use specimen in accordance with figure 1 of this specification. The thickness of the specimen shall be the full thickness of the blank provided that a small amount of metal shall be removed by machining when necessary to obtain a suitable test specimen. For blanks less than 0.5 inch in width, the manufacturer shall provide sample blanks of sufficient width to make the tension test specimens from the same material as the blanks represented. For blanks of inside diameter less than 1 inch a sample tube from which blanks are cut shall be tested in full section in accordance with Fed. Test Method Std. No. 151. In cases where the blanks are cut from hard tubes and subsequently annealed, the test section shall be annealed with the blank it represents.

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4.5.2 Test procedures.

4.5.2.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed in accordance with method 111 or method 112 of Fed. Test Method Std. No. 151 to determine conformance with 3.2. A single analysis of a composite sample may be made. In case of dispute, analysis in accordance with method 111 of Fed. Test Method Std. No. 151 shall be the basis for acceptance.

4.5.2.2 Tension tests. Tension tests shall be conducted in accordance with method 211 of Fed. Test Method Std. No. 151 to determine conformance with 3.3.

4.5.2.3 Bend tests.

4.5.2.3.1 Bend test specimens shall consist of a section of the full width and thickness of the band blank with the subtended arc not less than one-third nor more than one-half the circumference of the band blank. The specimen shall be bent cold, by pressure, without hammering, in the direction of curvature of the band blank, until it tightly incloses 180° of a rod equal in diameter to the thickness of the band blank.

4.5.2.3.2 As an alternate to the bend test, a full band blank shall be flattened so that a gage set at three times the wall thickness will pass over the tube freely throughout the flattened part. In making this test, the specimens will be flattened slowly by one stroke of the press.

4.5.2.3.3 Elements tested in accordance with 4.5.2.3.1 or 4.5.2.3.2 shall meet the bend test requirement of 3.4. In case of dispute, 4.5.2.3.1 shall be the basis for acceptance.

4.5.2.4 Grain size determination. Grain size shall be determined in accordance with method 312 of Fed. Test Method Std. No. 151 to determine conformance with 3.3.

4.6 Rejection.

4.6.1 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified for that sample size, the entire lot shall be rejected subject to the provisions of the section on "Acceptance and Rejection" of MIL-STD-105.

4.6.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.5 subject to the provisions of the section on "Rejection and Retests" of Fed. Test Method Std. No. 151.

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5. PREPARATION FOR DELIVERY

5.1 Packing. (See 6.2).

5.1.1 Levels A and B. The material shall be packed in accordance with MIL-C-3993.

5.1.2 Level C. The material shall be separated by size, composition and temper and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, in such a manner as to insure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

5.2 Marking.

5.2.1 In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Copper or copper alloy number and temper of the material required (1.2).
(c) Dimensions of blanks.
(d) Whether material is to be packed by level A, B, or C (see 5.1).
(e) Special marking, if required (see 5.2.1).

6.2 The requirements for packing and marking for shipment (see 5.1 and 5.2) specified herein apply to direct shipment for Government activities and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

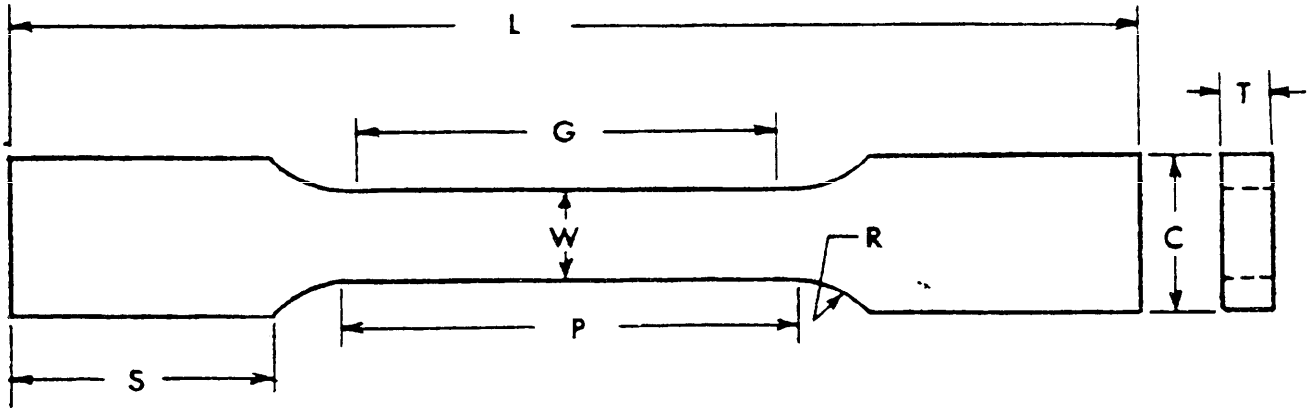
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Dimension of Test Specimen	Inside diameter (inches)	
	1.0 to 1.875 incl.	Over 1.875 to 7.5 excl.
	Inches	Inches
G (gage length)	1	2
L Approximate	(1)	9^{2}
P minimum	1.125	2.25
R (radius), minimum	0.5	0.5
C approximate	(3)	3^{2}
T4	(5)	(5)
W6	0.500 ± 0.010	0.500 ± 0.010
C7 approximate	5/8	5/8

Notes:

1 Maximum obtainable.
2 These dimensions shall be the maximum obtainable when the flattened blank is less than 9 inches long.
3 Remainder of blank.
4 See 4.5 for thickness requirements.
5 Band thickness.
6 See 4.5 regarding blanks less than 1/2 inch wide.
7 When blank is 1/2 inch to 5/8 inch, inclusive, in width, the full width of the blank shall be used.

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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